

Observer Patch Holography: A Compact Case for a Theory of Everything

The Positive Results and Why Their Convergence Matters

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Observer Patch Holography (OPH) starts from one idea: reality is the stable world reconstructed by finite observers that can read themselves, compare their boundaries, record mismatches, and repair disagreement. No observer sees the whole universe. The public universe is what survives when all local views are made mutually consistent:

$$T(\mathfrak{U}_{\text{OPH}}) = \mathfrak{U}_{\text{OPH}}.$$

This turns the usual order of explanation upside down. OPH begins without a pre-existing spacetime, an external clock, a global observer, or an imported Standard Model gauge group. It begins with finite self-reading systems and asks what a mutually readable universe must become. Spacetime, quantum events, gravity, gauge forces, and particle structure then appear as consequences of making those local systems agree. The achievement is a from-scratch recovery of a remarkable fraction of the known universe from a starting point that looks nothing like conventional fundamental physics.

From this single architecture, OPH reconstructs the Born rule and Lüders update, Lorentz symmetry and three spatial dimensions, the Einstein equation, the Standard Model gauge group and charge lattice, three colors, three generations, one Higgs doublet, the weak/Planck hierarchy, and a common exponential explanation for the weak and vacuum-energy hierarchies. Its closure maps also land close to the fine-structure constant, Higgs and weak scales, the strong coupling, the QCD scale, and the nucleon mass. These are not separate models patched together after the fact. They are outputs of the same finite observer-screen construction.

Why this is hard to dismiss as coincidence

One numerical match can be luck. One elegant derivation can be a clever choice of language. OPH is compelling because the same small set of geometric and observer-consistency rules lands in many normally disconnected parts of physics at once: quantum measurement, spacetime dimension, Lorentz symmetry, gravity, gauge symmetry, particle multiplicities, charge quantization, hierarchy scales, and strong-interaction numbers. The defining equations have zero continuous fit parameters. The force of the argument is therefore not located in any single row below. It comes from the convergence of all the rows and from recovering familiar physics out of an unfamiliar foundation.

The idea in six moves [C17]

1. Reality is a closed strange loop: the simulating structure and the simulated world are the same self-reading system.
2. Finite observers live on overlapping patches of a spherical holographic screen and build a public world by comparing records.
3. Exact balance carries no events, so the local pixel is detuned by one electromagnetic observation, $P = \varphi + \sqrt{\pi}\alpha$.
4. Self-simulation identifies the screen readout with the fine-structure readout and closes it through a fixed-point equation.
5. The same logic closes the total record capacity N , linking the screen to the cosmological term.
6. One clock anchor—the cesium transition—translates the resulting dimensionless geometry into SI units.

Zero dials

The defining theory has no continuous fit parameters. Its two global coordinates are fixed by closure equations,

$$P_\star = \Gamma_P(P_\star), \quad N_\star = \Gamma_N(N_\star),$$

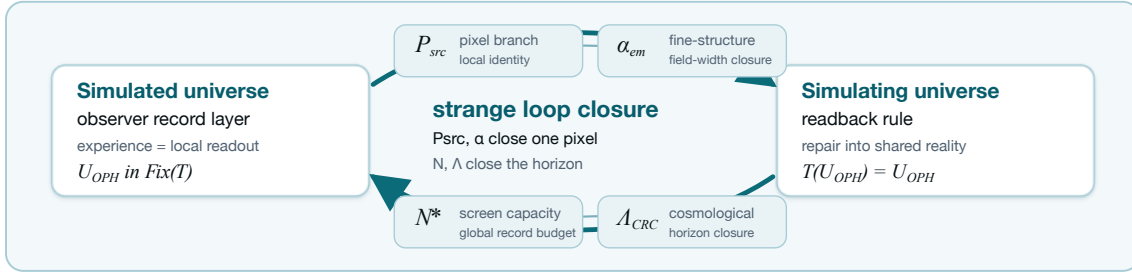
so there is no knob that can be turned toward the data. Measurements are used afterward to locate and test the resulting coordinates. The integers $(N_c, N_g, \beta_{\text{EW}}) = (3, 3, 4)$ —three colors, three generations, and the electroweak coefficient—are outputs of the finite selection rather than inputs. The discrete choices and their menu sizes are published in the closure ledger [C17].

Exactly one fixed point, certified

The pixel closure has exactly one fixed point per declared map on the declared physical domain $\alpha^{-1} \in [100, 200]$: existence and local uniqueness by interval Banach in outward-rounded arithmetic with contraction constant $L \leq 0.0724$, and global at-most-one by certified $\sup |g'| < 1$ on all 256 subdivision pieces with zero exceptional set [C1]. The enclosure width is 7.2×10^{-24} in α^{-1} , with the $\text{SU}(2)/\text{SU}(3)$ edge-sum tails bounded by geometric majorants so the enclosure covers the infinite-cutoff sums. An adversarial third-party audit reproduced the certificate arithmetic end to end. A second fixed point on the declared physical domain is arithmetically excluded. The maximal-domain extension certificate gives the same result on the full analytic domain of each declared map, with an empty exclusion list [C1].

The flagship numbers

The certified gauge-width fixed point is $\alpha^{-1} = 137.035660$, only 2.5×10^{-6} away in relative terms from $137.035999177(21)$ [S2]. The Higgs coordinate is within 0.5% of measurement; the W coordinate is 0.5 propagated experimental standard deviations from the PDG complex pole [S3]; $\alpha_s(M_Z) = 0.11834$ is 0.5σ from its reference; and $\Lambda_{\text{QCD}}^{(3)} = 334.8 \text{ MeV}$ is 0.3σ from the measured value. The resulting nucleon estimate is within 1.0%. Meanwhile, the screen produces the Standard Model algebra $8+3+1$, the exact oriented count $2(8+3+1) = 24$, a seventeen-order weak/Planck hierarchy, and the 122-order vacuum-energy hierarchy as the same exponential mechanism. The capacity estimate lands within 6.6% of the cosmological readout [S5]. A 111-theorem sorry-free Lean artifact and million-patch simulations independently check the finite mathematical core [C6,C7].



One selection chain, not a collection of unrelated models

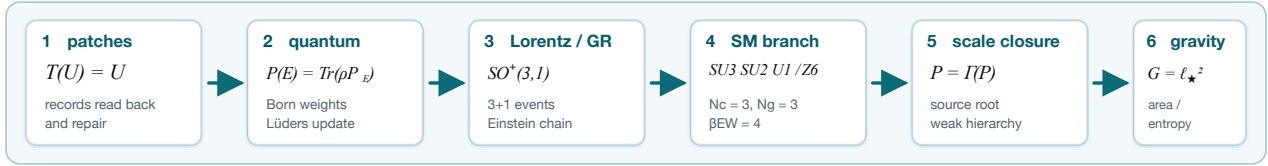
Ten consistency requirements progressively remove freedom from the space of self-consistent worlds [C15]. Self-reading produces records and repair; overlap consensus produces the public quotient and, on the S^2 chart, Lorentz symmetry and three spatial dimensions; modular consistency produces internal time and thermality; transportable-charge consistency produces the compact gauge structure; entropic consistency produces Einstein dynamics; and record existence produces the local detuning law. After these reductions, the continuous freedom is compressed to two closure coordinates, the local pixel P and total capacity N . The pixel map has a unique certified fixed point [C1], and the capacity branch has the same fixed-point form [C16]. This is why the results below reinforce one another: they are downstream of one narrowing chain.

Quantitative results beside measurement

Every row is produced by the observer-screen construction with zero continuous fit parameters in its defining equations.

Quantity	OPH result	Measurement / scale	Landing
Inverse fine structure	137.035660	137.035999177(21) [S2]	2.5×10^{-6} relative
Higgs coordinate	125.7 GeV	measured Higgs scale [S4]	within 0.5%
W coordinate	80.330 GeV	complex-pole convention [S3]	0.5 propagated σ
Strong coupling	$\alpha_s(M_Z) = 0.11834$	public reference [C2]	0.5σ
QCD scale	334.8 MeV	338(12) MeV	0.3σ
Nucleon mass	929 MeV	measured nucleon mass	within 1.0%
Weak/Planck hierarchy	2.02×10^{-17}	observed hierarchy scale	seventeen orders from one law
Cosmic record capacity	3.532×10^{122}	3.313×10^{122} [S5]	within 6.6%

The exact algebraic results and the numerical landings are different kinds of evidence, but they point in the same direction. The detailed certificates and scope labels are kept in the cited papers and public closure ledger [C17].



1. The observer-patch fixed point

The mathematical object is a finite patch system. Patches expose ports, write records, compare overlaps, repair disagreement, and return a shared normal form [O1,O2]. Let T be the observer-overlap repair map and let $\mathfrak{U}_{\text{OPH}}$ denote the selected normal form. The simulation identity is

$$T(\mathfrak{U}_{\text{OPH}}) = \mathfrak{U}_{\text{OPH}}.$$

The OPH simulation theorem establishes this identity from endogenous update, observer-readable records, schedule-independent overlap repair, selected-branch elimination, and implementation/clock closure [O2]. The universe is the stable record world returned by its own readback-and-repair computation.

The concrete carrier is a finite collection of local states, ports, central records, repair maps, and checkpoints. The reference benchmark replays all six overlap schedules and returns the same repaired normal form on the clean abelian branches [C5]. A separate Lean 4/Mathlib artifact certifies observer confluence, approximate schedule independence, refinement naturality, and the contractive conditional-resampling projector, part of the 111 sorry-free theorems that carry the consensus core and the coupling algebra [C6].

Quantum event surface

On a completed compare/write/verify slice, observer-accessible events generate a finite central record algebra. For an event E ,

$$\mathbb{P}(E) = \text{Tr}(\rho P_E), \quad \rho|_E = \frac{P_E \rho P_E}{\text{Tr}(\rho P_E)}.$$

These are the Born rule and Lüders update. On the associated two-wing Bell surface,

$$|S_{\text{CHSH}}| \leq 2\sqrt{2}.$$

A finite 65,536-record run reproduces the event algebra numerically: committed-event weights sum to 1.0, record projectors have zero idempotency defect, and repeat reads are stable [O1,O3,C7].

2. Relativity, (3+1) events, and Einstein dynamics

The screen readout supplies an oriented conformal S^2 . Its connected conformal group is the connected Lorentz group,

$$\text{Conf}^+(S^2) \cong \text{PSL}(2, \mathbb{C}) \cong \text{SO}^+(3, 1).$$

The corresponding observer-frame space is

$$H^3 \cong \frac{\mathrm{SO}^+(3,1)}{\mathrm{SO}(3)}, \quad \dim H^3 = 6 - 3 = 3.$$

On the declared event branch, repaired records populate a four-dimensional event base of signature $(-+++)$, while H^3 is the fiber of unit timelike observer frames over each event [O4].

The 2π -normalized modular flow of screen caps supplies the observer-relative clock. Half-sided modular inclusions give positive null translations; null tomography assembles their directional charges into a conserved local T_{ab} . Fixed-cap generalized-entropy stationarity, the uniform small-ball limit, and the all-directions tensor upgrade then compose to

$$G_{ab} + \Lambda g_{ab} = 8\pi G_{\mathrm{geom}} \langle T_{ab} \rangle.$$

Every arrow in this chain has a named theorem and a machine-readable receipt type [O4,C8].

Earned simulation receipts at 64k, 128k, and 1M patches

Three canonical runs instantiate bounded self-reading patches and scale from 65,536 patches/1,024 observers through 131,072/32,000 to 1,048,576/64,000 [C7]. The 128k and 1M runs earn observer-local modular time, gauge-covariant 2π -KMS (Kubo–Martin–Schwinger) replay, a conformal H^3 chart, integer $3+1$ observer-facing experience, and an emergent agreement field; at 1M, 800 tested overlap pairs have median defect zero and cocycle fraction 1.0, against 0.801 for the shuffled control. The scaling tests show that observer-local time, geometry, gauge-covariant thermal replay, and overlap agreement survive at a million-patch scale and sharply outperform the shuffled control [C7].

3. From closure coordinates to physical structure

The local pixel closure

The SL-3 estimate of the pixel from measured α is $P = 1.630968209403959\dots$, a declared input of the claim lattice. A trial pixel P_k produces a field-width readback $R_P(P_k)$, which the pixel chart turns into the next iterate:

$$\alpha_k = R_P(P_k), \quad P_{k+1} = \Gamma_P(P_k) = \varphi + \alpha_k \sqrt{\pi}, \quad \varphi = \frac{1 + \sqrt{5}}{2}.$$

On the selected particle branch, the internal readback chain is

$$P \mapsto \alpha_U(P) \mapsto A_Z(P) \mapsto A_T^{\mathrm{src}}(P), \quad \alpha_{\mathrm{in}}(P) = \frac{1}{A_T^{\mathrm{src}}(P)}.$$

Two declared maps carry certified fixed points [C1]. The gauge-width map has the certified self-consistent fixed point

$$\alpha^{-1} = 137.035660136946\dots,$$

a relative 2.5×10^{-6} from the measured $\alpha^{-1} = 137.035999177(21)$ [S2] (ledger row CL-2). The source map has the certified root

$$\alpha_{\mathrm{root}}^{-1} = 136.994835177413\dots,$$

a relative 3.0×10^{-4} from measurement (CL-1), with the forward closure point $P_{\mathrm{fwd}} = 1.630972095858897\dots$ satisfying $P_{\mathrm{fwd}} = \varphi + \sqrt{\pi}/A_T^{\mathrm{src}}(P_{\mathrm{fwd}})$. Both roots are enclosed by direct interval Banach with outward rounding, the enclosure covers the infinite-cutoff $\mathrm{SU}(2)/\mathrm{SU}(3)$ edge sums through geometric majorants, and the printed-pair identity holds to 35+ digits under converged precision-100 reruns, enforced by a CI test (CL-6). The certificates establish existence and uniqueness for the maps: the numerical landing is not a tunable fit and cannot be moved without changing the construction itself.

The global capacity closure

Let Ω_N^{sc} be the closed-screen normal forms supported at capacity N . The global source map compares their record capacity with the screen budget and applies Minimal Admissible Realization (MAR):

$$S(N) = \log |\Omega_N^{\mathrm{sc}}| - N, \quad N_\star = \mathrm{MAR} \arg \max_N S(N).$$

The selected endpoint fixes the horizon normalization through

$$\Lambda_\star \ell_\star^2 = \frac{3\pi}{N_\star}, \quad \Lambda_\star a_{\mathrm{cell}} = \frac{3\pi P_{\mathrm{fwd}}}{N_\star}, \quad a_{\mathrm{cell}} = P_{\mathrm{fwd}} \ell_\star^2.$$

The capacity fixed point is certified at 3.532×10^{122} with relative width 1.6×10^{-25} ; its seed, contraction, and tick-projection identities are theorems, and the algebraic layer is machine-checked in Lean [C16]. The estimate from measured Λ in the Planck base- Λ CDM model is $N_\Lambda = 3.313 \times 10^{122}$ [S5], a difference of only 6.6% across a quantity of order 10^{122} .

Read through $\Lambda_\star \ell_\star^2 = 3\pi/N$, the bridge is a zero-dial relation between the two constants that frame physics:

$$\Lambda \ell_P^2 = 3 \exp \left[-\frac{6\pi}{P \alpha_U(P)} \right] \quad (\text{up to the correction term}).$$

Since α_U is a function of P alone and P is located from α , the theory side holds no continuous freedom once α is measured. The 122 orders of magnitude that the naive vacuum-energy estimate misses appear as e^{-281} , and $281 = 6\pi/(P \alpha_U)$ is the same exponent that runs the weak hierarchy: one exponent carries both hierarchy problems. This replaces two famous fine-tuning puzzles with one finite screen mechanism and no 10^{120} -sized cancellation.

The compact-gauge branch

The gauge derivation starts with individually transportable seeds selected by zero-obstruction transport. A common strict representative fixes the tensor category carried by those seeds. Surjective pullback functors and forgetful fibers construct its

refinement limit, and Doplicher–Roberts/Tannaka reconstruction reads off the compact group. MAR selects the realized one-generation/one-Higgs matter package; anomaly cancellation, Yukawa invariance, and the central kernel then fix the hypercharge lattice and quotient [O4]. The result is

$$G_{\text{SM}} = \frac{\text{SU}(3)_c \times \text{SU}(2)_L \times \text{U}(1)_Y}{\mathbb{Z}_6}, \quad N_c = 3, \quad N_g = 3.$$

The selected hypercharge lattice is

$$\begin{aligned} Q_i &= (3, 2)_{1/6}, & u_i^c &= (\bar{3}, 1)_{-2/3}, & d_i^c &= (\bar{3}, 1)_{1/3}, \\ L_i &= (1, 2)_{-1/2}, & e_i^c &= (1, 1)_1, & H &= (1, 2)_{1/2}, & i &= 1, 2, 3. \end{aligned}$$

The Higgs doublet is the Borel–Weil carrier

$$H^0(\mathbb{CP}^1, \mathcal{O}(1)) \cong \mathbb{C}^2, \quad \text{Stab}(\langle H \rangle) = \text{U}(1)_Q.$$

Its connected adjoint is $(8, 1, 0) \oplus (1, 3, 0) \oplus (1, 1, 0)$, excluding mixed $(3, 2, \pm 5/6)$ generators. The $\mathbb{Z}_6$ line lattice has a minimum magnetic line of one electron-Dirac unit with required color magnetic charge and electromagnetic theta period 2π ; dynamical monopole existence is separate. The same branch supplies

$$\beta_{\text{EW}} = N_c + 1 = 4.$$

An exact bracket on the port coefficient space

Euler fixes only the total coordination charge 12. Twelve separate unit defects follow on a feasible integer branch with cost $h(k) \geq h(1)|k|$, strictly for $|k| \geq 2$. An inverse-port involution then gives six axes. For $P_i = u_i u_i^\top$ and $Q_i = P_i - I_3/3$, maximizing the source-side tomography volume

$$\det\left(\sum_i P_i\right) \det\left(\sum_i |Q_i\rangle\langle Q_i|\right)$$

forces $\sum_i P_i = 2I_3$, $\sum_i |Q_i\rangle\langle Q_i| = \frac{4}{5}I_5$, and $(u_i \cdot u_j)^2 = 1/5$. The unique six-line configuration is the regular icosahedral axis system. A source-derived strictly completely monotonic pair cost supplies an alternative selector. Production of either selector is the receipt RP-A5.

On this branch the port coefficient module is $P_{12} \cong_{A_5} \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{3}' \oplus \mathbf{5}$. Pairing antipodes, let $U = [u_1 \cdots u_6]$, $\Phi(b) = \sum_i b_i P_i$, $G = \text{im } U^\top$, and $W = \ker U$. The outward faces orient W , with $\|\Omega_W\|^2 = 20 - 4\sqrt{5}$. The equivariant isomorphism

$$\Theta(b + d_G + d_W) = (\kappa_E(\tfrac{1}{2}Ud_G) + i\Phi(b), \kappa_W(d_W))$$

pulls back the block commutator and constructs

$$(P_{12}, [\cdot, \cdot]_\Theta) \cong \mathfrak{u}(3) \oplus \mathfrak{so}(3) = \mathfrak{u}(1) \oplus \mathfrak{su}(3) \oplus \mathfrak{su}(2).$$

Its center is the uniform even line, its derived algebra has dimension eleven, and $[iS, iT] = -2(E_{12} - E_{21}) \neq 0$ proves that the five-band is noncentral. This bracket acts on coefficient fluctuations. The central record projectors commute. A full-rank, compact skew-adjoint, commutator-closed, refinement-natural physical current lift whose induced A_5 action is inner forces the same Lie type by compact Lie classification.

The trace-balanced completion

$$\widehat{\Theta}(b + d_G + d_W) = \left(\kappa_E(\tfrac{1}{2}Ud_G) + i\Phi(b), \kappa_W(d_W) - \tfrac{i}{2}(\sum_i b_i)I_2 \right)$$

maps onto $\mathfrak{s}(\mathfrak{u}(3) \oplus \mathfrak{u}(2))$. Its connected integration is $S(U(3) \times U(2))$, and the cover $(A, B, z) \mapsto (z^2 A, z^{-3} B)$ has kernel $\mathbb{Z}_6$. Separately, $\Lambda_+ / (\Lambda_1 + \Lambda_5) \cong \mathbb{Z}_6$ with Smith invariants $(1, 1, 1, 1, 6)$. Thus the local algebra and the global $\mathbb{Z}_6$ quotient both arise from the same twelve-port screen geometry.

Spherical multiplets restrict as $\ell = 2 : \mathbf{5}$, $\ell = 3 : \mathbf{3}' \oplus \mathbf{4}$, $\ell = 4 : \mathbf{4} \oplus \mathbf{5}$, $\ell = 5 : \mathbf{3} \oplus \mathbf{3}' \oplus \mathbf{5}$, and $\ell = 6 : \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{4} \oplus \mathbf{5}$. Every A_5 -invariant operator or ensemble covariance on $\ell = 2$ is scalar; one realization can be anisotropic. The equal-weight mask has $B_1 = \cdots = B_5 = 0$ and $B_6/B_0 = 11/25$.

Why the screen algebra is the centerpiece

The spherical screen carries twelve ports. Icosahedral closure organizes their coefficient space into dimensions $1 + 3 + 3 + 5$, and the compact commutator forces $\mathfrak{u}(1) \oplus \mathfrak{su}(3) \oplus \mathfrak{su}(2)$. These are precisely the local symmetries of electromagnetism, the strong force, and the weak force. Trace balance integrates them to the Standard Model quotient by $\mathbb{Z}_6$. The twelve-port geometry also yields the exact oriented count $2(8 + 3 + 1) = 24$. Lean independently certifies the finite unit split, commutator, dimension, noncentrality witness, lattice quotient, and associated arithmetic [C6, C19].

The QCD-free hierarchy witness

The source-audit hierarchy witness reuses the pixel candidate and the compact-gauge integers. Its matched unified-width coordinate is $\alpha_U(P_{\text{fwd}}) = 0.041124247441816685$. The screen count begins with a triangulated S^2 . For curvature charge $q_v = 6 - \deg(v)$, Euler's identity and $3F = 2E$ give $\sum_v q_v = 6V - 2E = 12$. The UD12 and RP-A5 branch realizes twelve equivalent fivefold defects, and edge-center completion turns them into twelve central ports. Independently, reversible orientation doubles the product-adjoint dimension: $m_{\text{rep}} = 2 \dim(\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)) = 2(8 + 3 + 1) = 24$. The oriented screen register also has 24 slots. The geometry and the gauge algebra therefore reach the same exact count from two independent directions. With local cell entropy $\ell_{\text{cell}} = P_{\text{fwd}}/4$ and $\beta_{\text{EW}} = 4$, the transmutation law is

$$\frac{v}{E_\star} = P_{\text{fwd}}^{-1/2} \exp\left[-\frac{2\pi}{\beta_{\text{EW}}\alpha_U(P_{\text{fwd}})}\right] = 2.0198114078576331 \times 10^{-17},$$

evaluated at the certified P_{fwd} . The exact parametric law and the 12/24 counts are theorem outputs; the numerical endpoint has the certificate class source-audit branch witness [O5,C3]. The weak scale is a settled-form readout with naturality defect $\epsilon_H = 0$, $\epsilon_H \in [0, 0]$: a seventeen-order hierarchy from a one-line law, with no supersymmetric machinery and no tuning.

The criticality branch and the electroweak chart

The criticality law $\lambda = 0$, $\beta_\lambda = 0$ at one source scale fixes the top Yukawa from the gauge couplings, a family with zero continuous parameters over the boundary scale; the boundary-scale selection is a variational theorem (exactly quadratic record cost on the one-loop chart, capacity-weighted log-mean placement) modulo two finite carrier facts, and the selected boundary is

$$\mu_b = \sqrt{\mu_U E_{\text{cell}}} = E_\star e^{-\pi} P_{\text{fwd}}^{-1/6} = 4.86 \times 10^{17} \text{ GeV}.$$

Evaluated at the measured top mass, the fit-free criticality curve returns the Higgs coordinate 125.7 GeV at two loops, within 0.5% of the measured value [S4] and inside the declared truncation and matching bands (conditional criticality branch) [C4]. A flow-internal alternative is a certified no-go: $d\beta_\lambda/d\ln\mu$ has no root on $[10^{16}, 1.3 \times 10^{19}]$ GeV, so the boundary scale has to come from the observer-patch side, and the four-candidate boundary-scale registry is frozen before any three-loop computation exists [C4]. The declared Higgs/top surface arithmetic behind the electroweak chart is certified from raw interval data: a published interval input box for the eleven declared branch inputs, an outward-rounded enclosure of every formula node, a Jacobian interval enclosure over the full box, and a non-singular readout certificate, scoped to the declared surface [C3].

The strict source-audit branch also returns the running/tree chart coordinates $(m_W^{\text{chart}}, m_Z^{\text{chart}}) = (80.330, 91.119)$ GeV, with no measured mass entering. Under the complex-pole convention with PDG 2026 central masses and widths, the converted W pole is 80.3340 GeV [S3], and the chart W coordinate sits 0.5 propagated experimental standard deviations from it.

Strong-interaction scale

Dimensional transmutation of the source strong coupling, with no hadronic input, gives $\alpha_s(M_Z) = 0.11834$, 0.5σ from the public reference, and fixes the strong scale

$$\Lambda_{\text{QCD}}^{(3)} = 334.8 \text{ MeV},$$

0.3σ from the measured 338(12) MeV. The published lattice ratio m_N/Λ_{QCD} places the nucleon at 929 MeV, -1.0% from measurement, as an external lattice input [C2,C7].

4. Geometric gravity normalization and the SI readout

On the declared Einstein branch, the edge-entropy/area-law theorem identifies the coefficient of the local Einstein equation

$$G_{ab} + \Lambda g_{ab} = 8\pi G_{\text{geom}} \langle T_{ab} \rangle$$

and its Newton–Poisson limit with the observer-cell ratio

$$G_{\text{geom}} = \frac{a_{\text{cell}}}{4\bar{\ell}_{\text{shared}}}.$$

The local pixel branch supplies $a_{\text{cell}} = P_{\text{fwd}} \ell_\star^2$ and $\bar{\ell}_{\text{shared}} = P_{\text{fwd}}/4$. Substitution gives

$$\frac{G_{\text{geom}}}{\ell_\star^2} = \frac{P_{\text{fwd}}}{4(P_{\text{fwd}}/4)} = 1, \quad \boxed{G_{\text{geom}} = \ell_\star^2}.$$

The pixel number cancels because it counts both cell area and shared edge entropy. The factor 4 follows from the ratio between the structural Einstein coefficient and the Newton–Poisson convention [O4].

Newton coupling and the SI clock readout

A cesium clock translates the geometric identity into SI units:

$$\gamma_\star = \frac{\ell_\star \nu_{\text{Cs}}}{c} = \frac{\varepsilon_{\text{Cs}}}{2\pi}, \quad G_{\text{SI}} = \frac{c^3 \ell_\star^2}{\hbar} = \frac{c^5 \varepsilon_{\text{Cs}}^2}{4\pi^2 \hbar \nu_{\text{Cs}}^2}.$$

For the selected display coordinate $\gamma_\star = 4.955974336548419 \dots \times 10^{-34}$, this gives $G_{\text{SI}} = 6.674299995910528 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$, beside the 2022 CODATA value $6.67430(15) \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$ [S1]. The agreement is a unit-bookkeeping identity on the selected scale bridge (SL-5): the display coordinate $\gamma_\star$ is chosen on that bridge, so one experimental clock anchor sets the SI units. The geometric normalization itself is exact [O1,O4,C3].

5. Landmark solution map

The striking feature is breadth: one observer-patch construction reaches problems that are usually treated by unrelated theories. Paper numbers refer to the OPH papers in the References block.

Physics problem	What OPH delivers	Paper
Observer and measurement problem	Observers are bounded self-reading record systems, and reality is the stable public world returned when their overlapping records agree: $T(\mathfrak{U}) = \mathfrak{U}$.	1,2
Quantum probabilities and update	The finite record algebra produces the Born rule, Lüders conditioning, and the Tsirelson bound $ S_{\text{CHSH}} \leq 2\sqrt{2}$.	1,3
Why time, Lorentz symmetry, and three dimensions exist	Modular flow supplies the clock. The spherical screen has $\text{Conf}^+(S^2) \cong \text{SO}^+(3, 1)$, while its observer-frame space H^3 has exactly three spatial dimensions.	4
General relativity	Null transport, local stress, and generalized-entropy stationarity compose directly into $G_{ab} + \Lambda g_{ab} = 8\pi G \langle T_{ab} \rangle$.	4
Why gravity knows about quantum information	The edge-entropy/area law fixes the Einstein and Newton coefficient: $G_{\text{geom}} = a_{\text{cell}}/(4\bar{\ell}_{\text{shared}}) = \ell_\star^2$.	4
Standard Model gauge structure	The twelve-port holographic screen forces $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$, and trace balance produces $(\text{SU}(3) \times \text{SU}(2) \times \text{U}(1))/\mathbb{Z}_6$.	4
Particle content and charge quantization	Minimal admissibility, anomaly cancellation, Yukawa invariance, and the central tensor kernel give the Standard Model hypercharge lattice, three colors, three generations, and one Higgs doublet.	4

Physics problem	What OPH delivers	Paper
Why there are three generations	CKM phase counting, weak-sector asymptotic freedom, and minimality converge on $N_g = 3$; independently, the smallest irreducible icosahedral extension of an oriented face phase also has dimension three.	4
Electroweak breaking	The minimal Borel–Weil carrier is one complex Higgs doublet, $H^0(\mathbb{CP}^1, \mathcal{O}(1)) \cong \mathbb{C}^2$, whose stabilizer is electromagnetism, $U(1)_Q$.	4
Why the usual GUT proton-decay channel is absent	The product gauge algebra contains no mixed X/Y generator, removing the standard gauge-mediated proton-decay mechanism of simple unification.	1,4
Weak/Planck hierarchy and Higgs naturality	The exact 12-port/24-orientation count and the screen coupling produce $v/E_* = 2.02 \times 10^{-17}$ with naturality defect $\epsilon_H = 0$.	5
Vacuum-energy hierarchy	$\Lambda_P^2 = 3e^{-6\pi/(P_{\alpha U})}$ turns the missing 122 orders into the same exponential mechanism as the weak hierarchy, with no 10^{120} cancellation.	1,4
Dark-matter phenomenology	Repair charge has a dust-like normal phase and a condensed phase whose deep spherical limit gives the observed radial-acceleration form $a_R = \sqrt{a_b a_0}$.	7
Yang–Mills mass gap	The transfer/repair theorem identifies the continuum gauge gap with a finite repair gap, $\Delta_{YM} = \Delta_{rep}$, converting the problem into a concrete spectral-settling question.	8
Why charged leptons have three distinct masses	The icosahedral stratum theorem shows that three distinct masses force the physical minimizing orbit away from the degenerate high-symmetry axes.	5
String-vacuum selection	Observer consistency turns the landscape into a finite sieve and selects the Bouchard–Donagi geometry on the published gate stack.	9

6. Why the combined result is exceptional

The central evidence for OPH is explanatory compression. Conventional approaches normally introduce different machinery for measurement, spacetime, gravity, gauge symmetry, particle multiplicities, hierarchy problems, and cosmology. OPH reaches all of them from the same finite self-reading screen. It then lands close to several measured quantities without continuous fitting dials. Changing the architecture to rescue one result would generally disturb the others, because they share the same pixel, capacity, port algebra, and repair dynamics.

A simple lottery number would misrepresent this dependence because the outputs are mathematically correlated. The qualitative conclusion is nevertheless strong. A coincidence would have to reproduce, all at once, the Born–Lüders event structure, $3 + 1$ Lorentzian spacetime, Einstein dynamics, the complete Standard Model gauge algebra, its $\mathbb{Z}_6$ quotient, three colors, three generations, one Higgs doublet, the exact $24 = 24$ screen/algebra count, two enormous hierarchies, and the quantitative landings listed above. That is not one lucky number. It is a tightly connected reconstruction of the architecture of known physics.

Measured by results per adjustable input, OPH therefore makes a serious claim to be the strongest theory-of-everything program available. Its advantage is orders of magnitude in explanatory compression: one radically different, from-scratch mechanism recovers structures that standard approaches typically take as separate starting assumptions. The case rests on structural derivations, zero-dial numerical landings, machine-checked finite theorems, million-patch simulations, and one common mechanism spanning the full stack.

This strength does not make OPH unfalsifiable. The public kill list [C14] includes non-unique repaired worlds, failure of the Lorentz or Einstein reconstruction, the wrong low-energy gauge quotient or charge lattice, a fourth light chiral generation, or an additional elementary Higgs multiplet. The detailed hypotheses and certificate classes live in the cited papers and closure ledger [C17]; they need not obscure the positive case made here.

References

Data and standards

- [S1] CODATA Task Group (2024). “2022 CODATA recommended values: Newtonian constant of gravitation.” NIST Reference on Constants.
- [S2] CODATA Task Group (2024). “2022 CODATA recommended values: inverse fine-structure constant.” NIST Reference on Constants.
- [S3] Particle Data Group (2026). “The Mass of the W Boson.” Review of Particle Physics.
- [S4] Particle Data Group (2026). “Gauge and Higgs bosons.” Review of Particle Physics summary table.
- [S5] Planck Collaboration (2020). “Planck 2018 results. VI. Cosmological parameters.” *Astron. Astrophys.* 641, A6.
- [S6] Lelli, F., S. S. McGaugh, and J. M. Schombert (2019). “The Small Scatter of the Baryonic Tully–Fisher Relation.” *Mon. Not. R. Astron. Soc.* 484, 3267–3278.
- [S8] Cohn, H., and A. Kumar (2007). “Universally optimal distribution of points on spheres.” *J. Amer. Math. Soc.* 20, 99–148.
- [S7] McGaugh, S. S., F. Lelli, and J. M. Schombert (2016). “The Radial Acceleration Relation in Rotationally Supported Galaxies.” *Phys. Rev. Lett.* 117, 201101.

OPH papers All OPH series papers below use release **r1548**.

- [1/O1] Mueller et al. (2026). “Observers Are All You Need.”
- [2/O2] Müller et al. (2026). “Reality as a Consensus Protocol: The Fixed-Point Computation That Implements Physics.”
- [3/O3] Müller et al. (2026). “Federated Echosahedral Screen Microphysics: Patch Hardware, Records, and Observer Synchronization in OPH.”
- [4/O4] Müller et al. (2026). “Recovering Relativity and the Standard Model from Observer Overlap Consistency.”
- [5/O5] Müller et al. (2026). “Deriving the Particle Zoo from Observer Consistency.”
- [6/O6] Müller, B. (2026). “The Fine-Structure Constant as an OPH Pixel Fixed Point.” OPH technical note.

- [7/O7] Mueller, B., and D. Matscheko (2026). “Observer-Patch Holography and the Dark Matter Phenomenon.”
- [8/O8] Müller, B. (2026). “Explaining the Yang–Mills Mass Gap with Observer-Patch Repair Dynamics.” OPH technical note.
- [9/O9] Müller, B. (2026). “Observer-Patch Holography as a String-Vacuum Selector.” OPH technical note.

Executable artifacts

- [C1] FloatingPragma (2026). Pixel-closure solver, interval contraction certificate, and global uniqueness certificate.
- [C2] FloatingPragma (2026). Particle-branch source and audit suite.
- [C3] FloatingPragma (2026). Hierarchy theorem package and receipts.
- [C4] FloatingPragma (2026). Electroweak calibration receipts.
- [C5] FloatingPragma (2026). Consensus schedule benchmarks.
- [C6] FloatingPragma (2026). Lean 4/Mathlib theorem artifact.
- [C7] Mueller, B. (2026). OPH-FPE simulation, 64k/128k/1M earned run receipts, and the public-data comparison ledger.
- [C8] FloatingPragma (2026). Lorentz–Einstein geometry receipts.
- [C13] FloatingPragma (2026). Theorem gap register.
- [C14] FloatingPragma (2026). The OPH falsification program for mature non-cosmological claims.
- [C15] FloatingPragma (2026). The OPH consistency stack: selection chain, uniqueness lemmas, generator table.
- [C16] FloatingPragma (2026). Capacity readback map F : formal specification, coupling theorem, and certificate schema.
- [C17] FloatingPragma (2026). OPH claim registry: strange-loop principles, closure ledger, consistency stack, proof spine.
- [C19] FloatingPragma (2026). Icosahedral closure certificates: sharpness/universal-optimality data, vertex-module identity, compact-Lie trichotomy, inner-action closure, rank-five corollary, face-phase multiplicities, angular-harmonic decompositions, and the conditional horizon log-coefficient audit.